CHAPTER 1
INTRODUCTION

1.1 Background

A job interview is one of the most important steps in a person's career. It is the point where a candidate has to present skills, knowledge and personality in a short time, and the result often depends as much on confidence and clarity of expression as on technical ability. Because of this, regular practice before an interview is widely considered essential, and many job seekers prepare with question banks, mock interviews and coaching sessions.

However, most interview-preparation resources are designed on the assumption that the candidate can hear and speak. Mock-interview tools usually ask questions aloud and expect spoken answers, and coaching is mostly given through conversation. For people who are Deaf, Hard of Hearing or Speech Impaired, this creates a real barrier. For many of them sign language is the most natural way to express ideas, and written language may be a second language. They therefore get very few opportunities to practise in a format that suits them, and they often enter real interviews with less preparation and less confidence than other candidates.

The widespread use of smartphones offers an opportunity to reduce this gap. Modern phones have good cameras that can record sign language, touchscreens that make typing easy, and constant internet access that allows data to be stored safely in the cloud. Cloud platforms now make it possible for a small team, or even a single developer, to build secure applications with user accounts, databases and file storage without managing servers.

InterviewSign is a mobile application developed to make interview practice accessible for Deaf, Hard of Hearing and Speech Impaired job seekers, while remaining open to anyone who prefers to answer in text or sign. A user creates an account with a unique username and a profile that includes a hearing or speech status and an answer preference of Text, Sign Video or Both. The user then selects an interview round such as HR Interview, Technical, Communication or Behavioral, chooses an option within that round, picks one or more topics, and practises by answering questions either by typing or by recording a sign language video. After answering, the user receives feedback and a score, and every attempt

is recorded so that progress can be reviewed through attempt history, average score and streak.

The application is built with React Native and Expo using TypeScript, and screens are organised with file-based routing through expo-router. Supabase provides email and password authentication, a PostgreSQL database for profiles and attempts, and storage for profile photographs. Application-wide state such as the logged-in user is handled through React Context. Since React Native produces a single codebase, the same application runs on both Android and iOS devices, which helps it reach as many users as possible.

1.2 Problem Statement

Deaf, Hard of Hearing and Speech Impaired job seekers do not have an accessible and structured platform on which they can practise interview questions in the way they actually communicate. Existing preparation tools depend heavily on spoken interaction, which leaves these candidates with limited practice, little useful feedback and no clear way to measure their improvement.

The major problems identified are:

1. Most interview-practice tools expect spoken answers and do not support sign language or text-first communication.
2. Candidates with hearing or speech impairment get very little realistic practice, which reduces confidence before real interviews.
3. Feedback on answers is usually not available instantly, and general question lists do not tell the candidate how well he or she performed.
4. Preparation material is scattered across websites and books and is not organised by interview round, company type or topic.
5. There is no simple way to track progress over time, such as past attempts, scores and consistency of practice.
6. Many existing applications require complicated sign-up processes and do not allow the user to choose how he or she wants to answer.

There is therefore a need for a single, easy-to-use mobile application that lets such candidates choose their preferred way of answering, practise questions round by round and

topic by topic, receive immediate feedback, and keep a record of their progress.

1.3 Objectives

The main aim of the project is to design and develop a mobile application that helps Deaf, Hard of Hearing and Speech Impaired people prepare confidently for job interviews. The specific objectives are listed below.

1. To develop a user-friendly, cross-platform mobile application for interview practice using React Native and Expo.
2. To provide secure registration and login using an email address or a unique username, with real-time checking of username availability.
3. To allow users to answer interview questions either by typing text or by recording a sign language video, according to their answer preference.
4. To organise practice into interview rounds, options and topics so that users can select exactly what they want to prepare, for example HR, Technical, Communication or Behavioral questions.
5. To give users instant feedback and a score after each answer so that they can understand their strengths and areas that need improvement.
6. To store every attempt and display it in a history with filters by category, answer type and colour-coded scores.
7. To show progress through statistics such as number of attempts, average score and practice streak on the home screen.
8. To let users manage a personal profile, including a profile photograph and editable details, with a controlled username change policy.
9. To protect user data by using authenticated access, protected navigation routes and row level security on the database.
10. To increase confidence and employability of hearing- and speech-impaired candidates by providing an inclusive and accessible practice environment.